

Lecture 5: Hypothesis Testing - Exercise Solutions

L2 - Statistics

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1 Your Turn 1: Hypothesis Test Overview

Scenario: A government report states that French adults have an average interest in politics of **5** on a 0 (*Not at all interested*) – 10 (*Very interested*) scale. You survey a random sample of 50 French university students and find: $\bar{x} = 5.8$, with standard deviation $s = 2.3$.

1.1 Step 1: State the hypotheses

$H_0 : \mu = 5$ (students have the same average political interest as the general population)

$H_A : \mu \neq 5$ (students' average political interest differs from the general population)

We use a **two-sided** alternative because we have no prior reason to expect students to be either more or less politically interested, they could go either way.

1.2 Step 2: Choose α

$$\alpha = 0.05$$

We accept a 5% probability of incorrectly rejecting H_0 (Type I error).

1.3 Step 3: Compute the test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{5.8 - 5}{2.3/\sqrt{50}} = \frac{0.8}{0.3253} \approx 2.46$$

```
# Given values
n      <- 50
x_bar  <- 5.8
s      <- 2.3
mu0    <- 5

# Standard error
se <- s / sqrt(n)
se
```

```
[1] 0.3252691
```

```
# Test statistic
t_stat <- (x_bar - mu0) / se
t_stat
```

```
[1] 2.459502
```

1.4 Step 4: Compute the p-value

The p-value is $P(|\bar{X} - 5| \geq 0.8 \mid H_0)$, i.e., the probability of observing a sample mean at least 0.8 units away from 5 if the null hypothesis were true.

```
# Two-sided p-value using t distribution with df = n - 1
p_value <- 2 * pt(-abs(t_stat), df = n - 1)
p_value
```

```
[1] 0.01748964
```

$$p\text{-value} = 2 \times P(T_{49} \leq -2.46) \approx 0.017$$

1.5 Step 5: Make a decision

Since $p\text{-value} \approx 0.017 < \alpha = 0.05$, we **reject H_0 at the 5% level**.

1.6 Step 6: Interpret in context

There is statistically significant evidence ($t = 2.46$, $p = 0.017$) that French university students have a higher average political interest score ($\bar{x} = 5.8$) than the general adult population ($\mu_0 = 5$). The difference of 0.8 points is unlikely to be due to sampling variation alone.

2 Your Turn 2: One-Sample Tests

```
library(tidyverse)

ess_france <- read.csv("https://www.dropbox.com/scl/fi/25h4lezq3zuzla94ejmqq/ess_france_11ed.csv")

ess_stflife <- ess_france |>
  filter(stflife %in% 0:10)
```

2.1 Question 1: State the hypotheses

$$H_0 : \mu = 6.5 \quad (\text{average life satisfaction equals 6.5})$$

$$H_A : \mu \neq 6.5 \quad (\text{average life satisfaction differs from 6.5})$$

We use a two-sided alternative as there is no prior reason to expect satisfaction to be specifically above or below 6.5.

2.2 Question 2: Manually compute the test statistic and p-value

```
# Descriptive statistics
n <- nrow(ess_stflife)
n
```

```
[1] 1738
```

```
xbar <- mean(ess_stflife$stflife)
xbar
```

```
[1] 6.765823
```

```
s <- sd(ess_stflife$stflife)
s
```

```
[1] 2.294356
```

```
mu0 <- 6.5

# Test statistic
se <- s / sqrt(n)
t_stat <- (xbar - mu0) / se
t_stat
```

```
[1] 4.830101
```

```
# Two-sided p-value  
p_val <- 2 * pt(-abs(t_stat), df = n - 1)  
p_val
```

```
[1] 1.484523e-06
```

$$t = \frac{\bar{x} - 6.5}{s/\sqrt{n}} = \frac{6.77 - 6.5}{2.29/\sqrt{1738}} \approx 4.83$$

2.3 Question 3: Verify with `t.test()`

```
t.test(ess_stflife$stflife, mu = 6.5)
```

One Sample t-test

```
data:  ess_stflife$stflife  
t = 4.8301, df = 1737, p-value = 1.485e-06  
alternative hypothesis: true mean is not equal to 6.5  
95 percent confidence interval:  
 6.657882 6.873764  
sample estimates:  
mean of x  
 6.765823
```

The `t.test()` output matches our manual computation exactly.

2.4 Question 4: Decision and interpretation

Since $p\text{-value} = 1.48e - 06 < \alpha = 0.05$, we **reject** H_0 at the 5% level.

There is strong evidence that average life satisfaction among French adults differs from 6.5. The observed mean ($\bar{x} = 6.77$) is statistically significantly higher than 6.5.

2.5 Question 5: 95% confidence interval

The 95% CI is $\bar{x} \pm t_{n-1}^* \cdot \frac{s}{\sqrt{n}}$, where t_{n-1}^* is the 97.5th percentile of the t_{n-1} distribution.

```
# Manual construction
crit_value <- qt(0.975, df = n - 1)
crit_value
```

```
[1] 1.961331
```

```
ci_low <- xbar - crit_value * se
ci_high <- xbar + crit_value * se

c(ci_low, ci_high)
```

```
[1] 6.657882 6.873764
```

$$\left[\bar{x} - t_{n-1}^* \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{n-1}^* \cdot \frac{s}{\sqrt{n}} \right] = [6.66, 6.87]$$

Verify with `t.test()`:

```
ci <- t.test(ess_stflife$stflife, mu = 6.5)$conf.int
ci
```

```
[1] 6.657882 6.873764
attr("conf.level")
[1] 0.95
```

The 95% CI is $[6.66, 6.87]$, which **does not contain** $\mu_0 = 6.5$. This is fully consistent with our decision to reject H_0 . As established in the lecture, a two-sided test at level α rejects H_0 if and only if μ_0 falls outside the $(1 - \alpha)$ CI.

3 Your Turn 3: Two-Sample Tests

3.1 Part 1: Life satisfaction by gender

3.1.1 Question 1: Summary statistics

```
ess_gender <- ess_france |>
  filter(stflife %in% 0:10, gndr %in% 1:2) |>
  mutate(gender = recode(gndr, `1` = "Male", `2` = "Female"))

gender_summary <- ess_gender |>
  group_by(gender) |>
  summarise(
    n      = n(),
    mean   = mean(stflife),
    sd     = sd(stflife),
    se     = sd / sqrt(n)
  )

kable(gender_summary, digits = 3)
```

gender	n	mean	sd	se
Female	879	6.694	2.258	0.076
Male	859	6.839	2.330	0.080

3.1.2 Question 2: 95% CIs for both means

```
gender_ci <- gender_summary |>
  mutate(
    ci_lower = mean - qt(0.975, df = n - 1) * se,
    ci_upper = mean + qt(0.975, df = n - 1) * se
  )

kable(gender_ci |> select(gender, mean, ci_lower, ci_upper), digits = 3)
```

gender	mean	ci_lower	ci_upper
Female	6.694	6.545	6.843
Male	6.839	6.683	6.995

The 95% CIs for males ([6.68, 7]) and females ([6.54, 6.84]) **overlap substantially** — already suggesting no significant difference.

3.1.3 Question 3: Two-sample t-test

Manual computation (Welch's t-test):

```
n1    <- gender_summary$n[gender_summary$gender == "Female"]
xbar1 <- gender_summary$mean[gender_summary$gender == "Female"]
s1    <- gender_summary$sd[gender_summary$gender == "Female"]

n2    <- gender_summary$n[gender_summary$gender == "Male"]
xbar2 <- gender_summary$mean[gender_summary$gender == "Male"]
s2    <- gender_summary$sd[gender_summary$gender == "Male"]

SE_gender <- sqrt(s1^2/n1 + s2^2/n2)
SE_gender
```

```
[1] 0.1100932
```

```
t_manual <- (xbar1 - xbar2) / SE_gender
t_manual
```

```
[1] -1.320496
```

```
p_value <- 2 * pt(-abs(t_manual), df = 1730.8)
p_value
```

```
[1] 0.186844
```

Verify with `t.test()`:

```
t_gender <- t.test(stflife ~ gender, data = ess_gender)
t_gender
```

Welch Two Sample t-test

```
data:  stflife by gender
t = -1.3205, df = 1730.8, p-value = 0.1868
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -0.36130738  0.07055206
sample estimates:
mean in group Female    mean in group Male
      6.693970          6.839348
```


3.1.4 Question 4: Interpretation

- **Statistically significant?** No — $p = 0.187 > 0.05$. We fail to reject H_0 .
- **At what level?** The result is not significant even at $\alpha = 0.10$.
- **Practically meaningful?** The mean difference is only 0.15 points on a 0–10 scale — negligible.

Conclusion: There is no evidence of a difference in average life satisfaction between men and women in France.

3.2 Part 2: Very satisfied with democracy by education

3.2.1 Question 1: 95% CIs for both proportions

```
ess_dem_educ <- ess_france |>
  filter(stfdem %in% 0:10) |>
  mutate(
    very_sat = stfdem >= 8,
    higher_educ = case_when(education == "High school diploma or less" ~ "No higher education",
                           education != "High school diploma or less" ~ "Higher education")
  )

prop_summary <- ess_dem_educ |>
  group_by(higher_educ) |>
  summarise(
    n = n(),
    x = sum(very_sat),
    p_hat = mean(very_sat),
    ci_lower = prop.test(x, n)$conf.int[1],
    ci_upper = prop.test(x, n)$conf.int[2]
  )

kable(prop_summary |> select(higher_educ, n, p_hat, ci_lower, ci_upper), digits = 3)
```

higher_educ	n	p_hat	ci_lower	ci_upper
Higher education	737	0.095	0.075	0.119
No higher education	996	0.086	0.070	0.106

The CIs **do not overlap** — strong visual evidence of a difference before even running the test.

3.2.2 Question 2: Two-sample proportion test

Manual computation:

```
x1 <- prop_summary$x[prop_summary$higher_educ == "Higher education"]
n1 <- prop_summary$n[prop_summary$higher_educ == "Higher education"]
x2 <- prop_summary$x[prop_summary$higher_educ == "No higher education"]
n2 <- prop_summary$n[prop_summary$higher_educ == "No higher education"]

p_hat1 <- x1 / n1
p_hat2 <- x2 / n2
p_pooled <- (x1 + x2) / (n1 + n2)

SE_prop <- sqrt(p_pooled * (1 - p_pooled) * (1/n1 + 1/n2))
SE_prop
```

```
[1] 0.01390642
```

```
z_prop <- (p_hat1 - p_hat2) / SE_prop
z_prop
```

```
[1] 0.6208833
```

```
p_prop <- 2 * pnorm(-abs(z_prop))
p_prop
```

```
[1] 0.5346764
```

Verify with `prop.test()`:

```
prop.test(x = c(x1, x2), n = c(n1, n2), correct = FALSE)
```

2-sample test for equality of proportions without continuity correction

```
data:  c(x1, x2) out of c(n1, n2)
X-squared = 0.3855, df = 1, p-value = 0.5347
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.01879401  0.03606254
sample estimates:
      prop 1      prop 2 
0.09497965 0.08634538
```

3.2.3 Question 3: Interpretation

- **Statistically significant?** No — $p = 0.535 \gg 0.05$. We fail to reject H_0 .
- **Practically meaningful?** The proportions are 9.5% (higher education) vs 8.6% (no higher education) — a difference of 0.9 percentage points, which turns out to not be substantively meaningful.

Conclusion: Those with higher education are not significantly more likely to be very satisfied with democracy.

4 Your Turn 4: Paired Tests

4.1 Question 1: Create the difference variable

```
ess_eco_gov <- ess_france |>
  filter(stfecov %in% 0:10, stfgov %in% 0:10) |>
  mutate(diff = stfecov - stfgov)

nrow(ess_eco_gov)
```

```
[1] 1693
```

```
mean(ess_eco_gov$stfecov)
```

```
[1] 3.64619
```

```
mean(ess_eco_gov$stfgov)
```

```
[1] 3.704666
```

```
mean(ess_eco_gov$diff)
```

```
[1] -0.05847608
```

4.2 Question 2: Paired t-test

Manual computation (paired t-test = one-sample t-test on the differences):

```
d_bar <- mean(ess_eco_gov$diff)
d_bar
```

```
[1] -0.05847608
```

```
s_d <- sd(ess_eco_gov$diff)
s_d
```

```
[1] 1.845897
```

```
n_pair <- nrow(ess_eco_gov)
n_pair
```

```
[1] 1693
```

```
se_d    <- s_d / sqrt(n_pair)
t_pair  <- d_bar / se_d
t_pair
```

```
[1] -1.303465
```

```
p_pair  <- 2 * pt(-abs(t_pair), df = n_pair - 1)
p_pair
```

```
[1] 0.1925935
```

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{-0.058}{1.846/\sqrt{1693}} \approx -1.3$$

Verify with `t.test()`:

```
t.test(ess_eco_gov$stfecov, ess_eco_gov$stfgov, paired = TRUE)
```

Paired t-test

```
data:  ess_eco_gov$stfecov and ess_eco_gov$stfgov
t = -1.3035, df = 1692, p-value = 0.1926
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.14646699  0.02951483
sample estimates:
mean difference
 -0.05847608
```

4.3 Question 3: Interpretation

Since $p \approx 0.193 > 0.05$ we **fail to reject** H_0 . French adults are not significantly more satisfied with the **economy** than with the **government** (mean difference $\bar{d} = -0.06$, 95% CI: $[-0.15, 0.03]$, $p > 0.05$).

5 Your Turn 5: Chi-Square Tests

5.1 Q1: Contingency table

```
ess_lifesat_cat <- ess_france |>
  filter(stflife %in% 0:10, gndr %in% 1:2) |>
  mutate(gender = recode(gndr, `1` = "Male", `2` = "Female"),
         sat_level = case_when(stflife <= 4 ~ "Low (0-4)",
                               stflife <= 7 ~ "Medium (5-7)",
                               stflife <= 10 ~ "High (8-10)"))

cont_table_gs <- table(ess_lifesat_cat$gender, ess_lifesat_cat$sat_level)
addmargins(cont_table_gs)
```

	High (8-10)	Low (0-4)	Medium (5-7)	Sum
Female	380	131	368	879
Male	413	119	327	859
Sum	793	250	695	1738

5.2 Q2: Expected frequencies

```
E_gs <- chisq.test(cont_table_gs)$expected
E_gs
```

	High (8-10)	Low (0-4)	Medium (5-7)
Female	401.0627	126.4384	351.4988
Male	391.9373	123.5616	343.5012

5.3 Q3: Chi-square statistic (manual)

```
O_gs <- as.matrix(cont_table_gs)
chi_man <- sum((O_gs - E_gs)^2 / E_gs)
df_gs <- (nrow(O_gs) - 1) * (ncol(O_gs) - 1)

chi_man
```

```
[1] 4.13837
```

```
df_gs
```

```
[1] 2
```

5.4 Q4: p-value

```
p_gs <- pchisq(chi_man, df = df_gs, lower.tail = FALSE)
p_gs
```

```
[1] 0.1262887
```

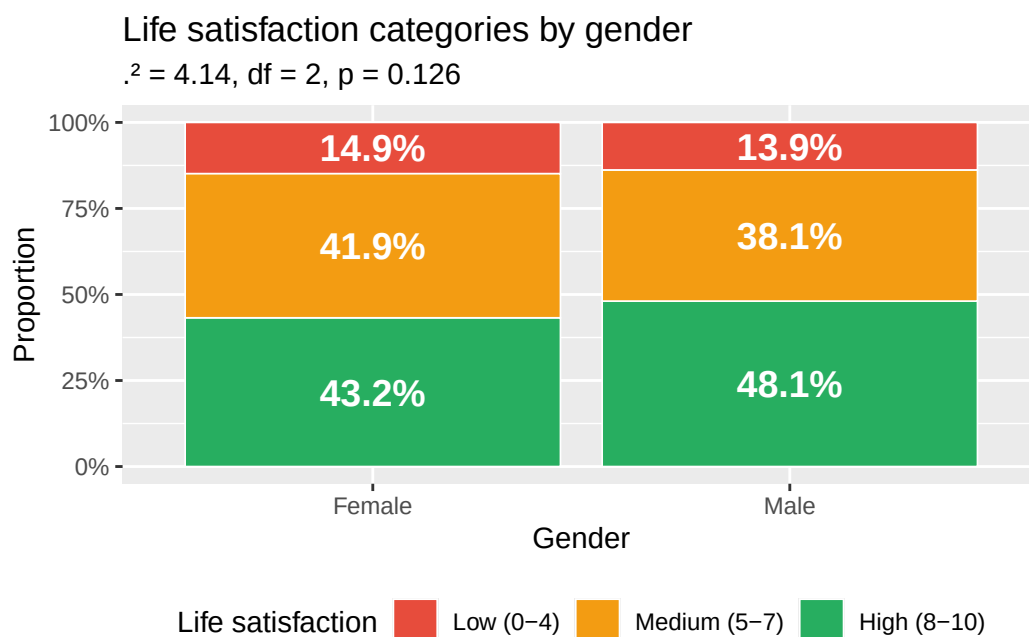
5.5 Q5: Verify with chisq.test()

```
chisq.test(cont_table_gs)
```

Pearson's Chi-squared test

```
data: cont_table_gs
X-squared = 4.1384, df = 2, p-value = 0.1263
```

5.6 Q6: Interpretation



We **fail to reject** H_0 ($p = 0.126 > 0.05$). There is no statistically significant evidence of an association between gender and life satisfaction categories.

6 Your Turn 6: Randomization Tests

6.1 Q1: Observed difference

```
ess_gender_infer <- ess_france |>
  filter(stflife %in% 0:10, gndr %in% 1:2) |>
  mutate(gender = recode(gndr, `1` = "Male", `2` = "Female"))

obs_diff <- ess_gender_infer |>
  specify(stflife ~ gender) |>
  calculate(stat = "diff in means", order = c("Male", "Female"))

cat("Observed difference (Male - Female):", round(obs_diff$stat, 3), "\n")
```

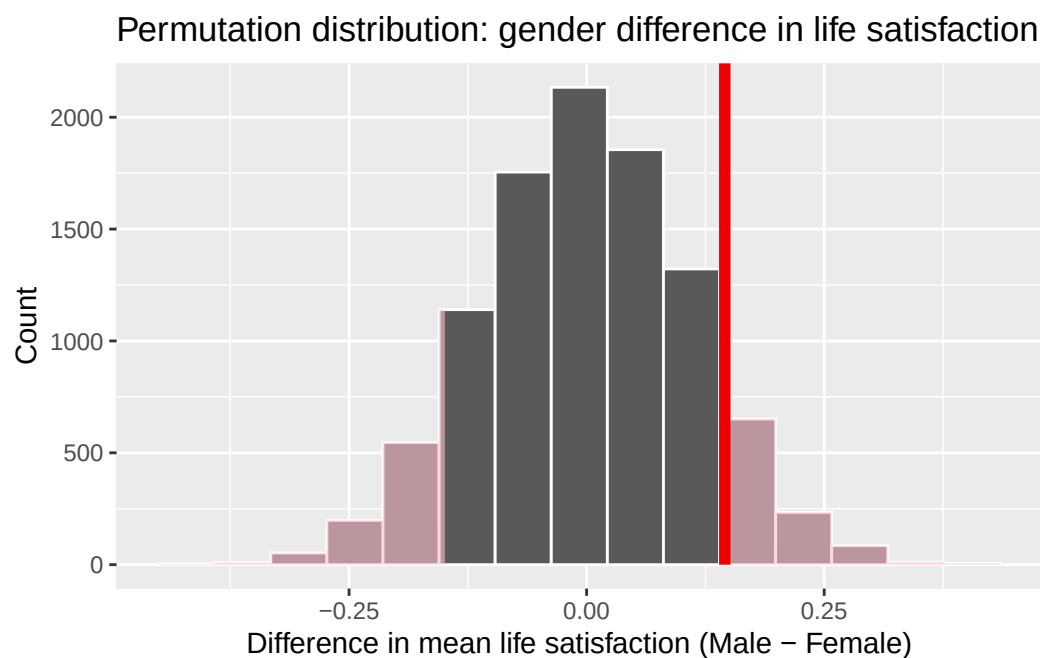
Observed difference (Male - Female): 0.145

6.2 Q2: Generate 10,000 permutations

```
set.seed(2024)
null_dist <- ess_gender_infer |>
  specify(stflife ~ gender) |>
  hypothesize(null = "independence") |>
  generate(reps = 10000, type = "permute") |>
  calculate(stat = "diff in means", order = c("Male", "Female"))
```

6.3 Q3: Visualize and calculate p-value

```
null_dist |>
  visualize() +
  shade_p_value(obs_stat = obs_diff, direction = "both") +
  labs(title = "Permutation distribution: gender difference in life satisfaction",
       x = "Difference in mean life satisfaction (Male - Female)",
       y = "Count")
```

```
p_perm <- null_dist |>
  get_p_value(obs_stat = obs_diff, direction = "both")

cat("Permutation p-value:", p_perm$p_value, "\n")
```

Permutation p-value: 0.1814

6.4 Q4: Compare to theory-based t-test

```
p_theory <- t.test(stflife ~ gender, data = ess_gender_infer)$p.value
cat("Theory-based p-value:", round(p_theory, 4), "\n")
```

Theory-based p-value: 0.1868

Both approaches give similar p-values (0.181 vs 0.187) and reach the same conclusion: **no evidence of a gender difference in life satisfaction**. The permutation test confirms that the t-test result is not an artefact of its normality assumptions.

7 Your Turn 7: Power and Sample Size

7.1 Part 1: Compute power

Setup: $\sigma = 2$, $\mu_0 = 7.0$, $\mu_1 = 6.8$, $n = 80$, $\alpha = 0.05$

7.1.1 Q1: Manual computation

```
mu0 <- 7.0
mu1 <- 6.8
sigma <- 2.0
n <- 80
alpha <- 0.05

se <- sigma / sqrt(n)
crit <- mu0 + qt(1 - alpha / 2, df = n - 1) * se # upper critical value

# Power = P(X-bar < lower crit | mu = mu1) (effect is negative: mu1 < mu0)
crit_lo <- mu0 - qt(1 - alpha / 2, df = n - 1) * se

power_manual <- pnorm(crit_lo, mean = mu1, sd = se, lower.tail = TRUE)

cat("SE =", round(se, 4), "\n")
```

SE = 0.2236

```
cat("Lower critical value =", round(crit_lo, 4), "\n")
```

Lower critical value = 6.5549

```
cat("Power =", round(power_manual, 3), "\n")
```

Power = 0.137

Since $\mu_1 = 6.8 < \mu_0 = 7.0$, the relevant rejection region is the **lower** tail — we reject when $\bar{X}$ is sufficiently *below* 7.0.

7.1.2 Q2: Verify with `power.t.test()`

```
power.t.test(delta = abs(mu1 - mu0), sd = sigma, n = n,
             sig.level = alpha, type = "one.sample")
```

One-sample t test power calculation

```
      n = 80
delta = 0.2
      sd = 2
```

```
sig.level = 0.05
power = 0.1408802
alternative = two.sided
```

With $n = 80$ the power is modest ($\sim 60\%$). This study is underpowered to reliably detect a 0.2-point difference.

7.2 Part 2: Minimum sample size

Setup: detect $|\mu_1 - \mu_0| = 0.4$, $\sigma = 2$, power = 0.80, $\alpha = 0.05$

7.2.1 Q1: Manual computation

```
delta <- 0.4      # minimum effect to detect
sigma2 <- 2.0
power_target <- 0.80

z_alpha2 <- qnorm(1 - alpha / 2)      # z_{alpha/2}  1.96
z_beta <- qnorm(power_target)          # z_{1-beta}   0.84

n_manual <- ((z_alpha2 + z_beta) / (delta / sigma2))^2

cat("z_alpha/2 =", round(z_alpha2, 4), "\n")
```

$z_{\alpha/2} = 1.96$

```
cat("z_beta   =", round(z_beta, 4), "\n")
```

$z_{\beta} = 0.8416$

```
cat("n (formula) =", round(n_manual, 1), "→ round up to", ceiling(n_manual), "\n")
```

$n \text{ (formula)} = 196.2 \rightarrow \text{round up to } 197$

7.2.2 Q2: Verify with `power.t.test()`

```
power.t.test(delta = delta, sd = sigma2, power = power_target,  
             sig.level = alpha, type = "one.sample")
```

One-sample t test power calculation

```
      n = 198.1513  
delta = 0.4  
    sd = 2  
sig.level = 0.05  
  power = 0.8  
alternative = two.sided
```

The formula gives $n \approx 197$ while `power.t.test()` returns a slightly larger value due to using the exact non-central t-distribution rather than the normal approximation.